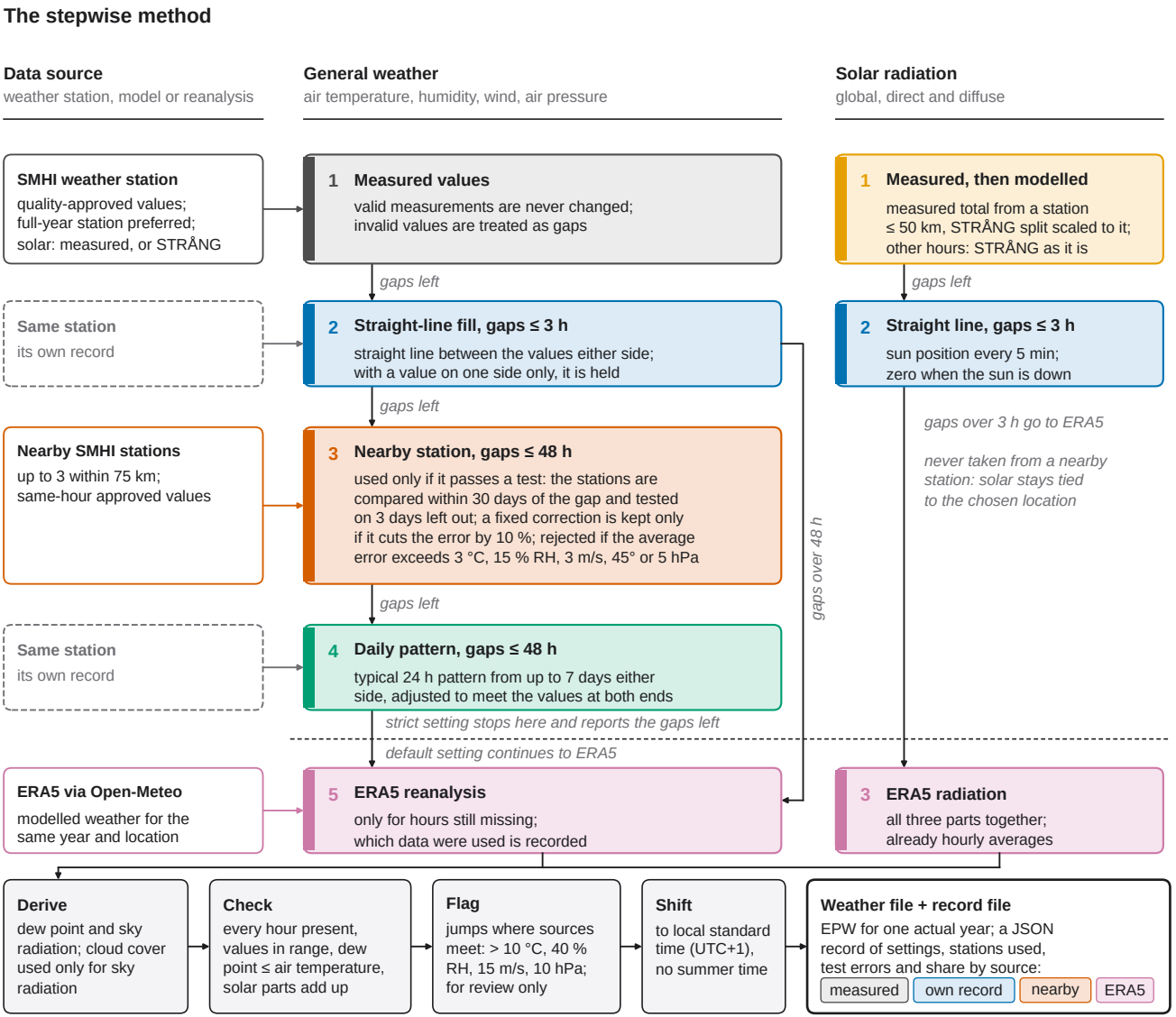
The stepwise method
Data source
weather station, model or reanalysis
General weather
air temperature, humidity, wind, air pressure
Solar radiation
global, direct and diffuse
SMHI weather station
quality-approved values;
full-year station preferred;
solar: measured, or STRÅNG
1 Measured values
valid measurements are never changed;
invalid values are treated as gaps
1 Measured, then modelled
measured total from a station
≤ 50 km, STRÅNG split scaled to it;
other hours: STRÅNG as it is
gaps left
gaps left
Same station
its own record
2 Straight-line fill, gaps ≤ 3 h
straight line between the values either side;
with a value on one side only, it is held
2 Straight line, gaps ≤ 3 h
sun position every 5 min;
zero when the sun is down
gaps left
gaps over 3 h go to ERA5
never taken from a nearby station: solar stays tied to the chosen location
Nearby SMHI stations
up to 3 within 75 km;
same-hour approved values
3 Nearby station, gaps ≤ 48 h
used only if it passes a test: the stations are compared within 30 days of the gap and tested on 3 days left out; a fixed correction is kept only if it cuts the error by 10 %; rejected if the average error exceeds 3 °C, 15 % RH, 3 m/s, 45° or 5 hPa
gaps over 48 h
gaps left
Same station
its own record
4 Daily pattern, gaps ≤ 48 h
typical 24 h pattern from up to 7 days either side, adjusted to meet the values at both ends
strict setting stops here and reports the gaps left
default setting continues to ERA5
ERA5 via Open-Meteo
modelled weather for the same year and location
5 ERA5 reanalysis
only for hours still missing;
which data were used is recorded
3 ERA5 radiation
all three parts together;
already hourly averages
Derive
dew point and sky radiation; cloud cover used only for sky radiation
Check
every hour present, values in range, dew point ≤ air temperature, solar parts add up
Flag
jumps where sources meet: > 10 °C, 40 % RH, 15 m/s, 10 hPa; for review only
Shift
to local standard time (UTC+1), no summer time
Weather file + record file
EPW for one actual year; a JSON record of settings, stations used, test errors and share by source:
measured
own record
nearby
ERA5